# *University of Information Technology & Sciences*

Department of Computer Science & Engineering

BSC in CSE- 3rd Year ,1st semester (Spring 2026)

CSE0611323: Microprocessors, Microcontrollers and Assembly Language

Section: 6B1

## Course Information

| Item | Details |
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| Course Code | CSE0611323 |
| Course Title | Microprocessors, Microcontrollers and Assembly Language |
| Section | 6B1 |
| Assignment-3 Title | **Perform Operations and fill up Observation Table** |
| Submission Date | 30 August 2026 |
| Course Teacher | Tahmina Yasmin Lima, Lecturer |

## Student Information

| Information | Details |
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| Section | 6B1 |

## Introduction:

In this Assignment-3, various logical, shift, and rotate operations are performed on 8-bit data using 8086 **microprocessor** Assembly Language. Bitwise operations work directly on the individual bits of binary data and are commonly used in low-level programming, system programming, and microprocessor-based applications. In this experiment, the **AND, OR, Shift Left (SHL), Shift Right (SHR), Rotate Left (ROL), and Rotate Right (ROR)** instructions were implemented using 8086 assembly language. Records and verifies the hexadecimal results of each operation through an observation table.

## CODE:

```
.model small
.stack 100h

.data
   and_result db ?
   or_result  db ?
   shl_result db ?
   shr_result db ?
   rol_result db ?
   ror_result db ?

.code
main proc

   mov ax, @data
   mov ds, ax

   mov al, 18h
   and al, 0Fh
   mov and_result, al

   mov al, 2Ah
   or al, 0D1h
   mov or_result, al

   mov al, 0C4h
   mov cl, 3
   shl al, cl
   mov shl_result, al

   mov al, 6Bh
```

```
mov cl, 6
shr al, cl
mov shr_result, al

mov al, 0FFh
mov cl, 2
rol al, cl
mov rol_result, al

mov al, 7Ah
mov cl, 7
ror al, cl
mov ror_result, al

mov dl, 13
mov ah, 2
int 21h

mov dl, 10
int 21h

mov al, and_result
call print_hex

mov dl, 32
mov ah, 2
int 21h

mov al, or_result
call print_hex

mov dl, 32
mov ah, 2
int 21h

mov al, shl_result
call print_hex

mov dl, 32
mov ah, 2
int 21h

mov al, shr_result
call print_hex
```

```
    mov dl, 32
    mov ah, 2
    int 21h

    mov al, rol_result
    call print_hex

    mov dl, 32
    mov ah, 2
    int 21h

    mov al, ror_result
    call print_hex

    mov ah, 4ch
    int 21h

main endp

print_hex proc

    mov bl, al

    mov cl, 4
    shr al, cl

    cmp al, 9
    jbe digit1
    add al, 7

digit1:
    add al, 30h
    mov dl, al
    mov ah, 2
    int 21h

    mov al, bl
    and al, 0Fh

    cmp al, 9
    jbe digit2
    add al, 7
```

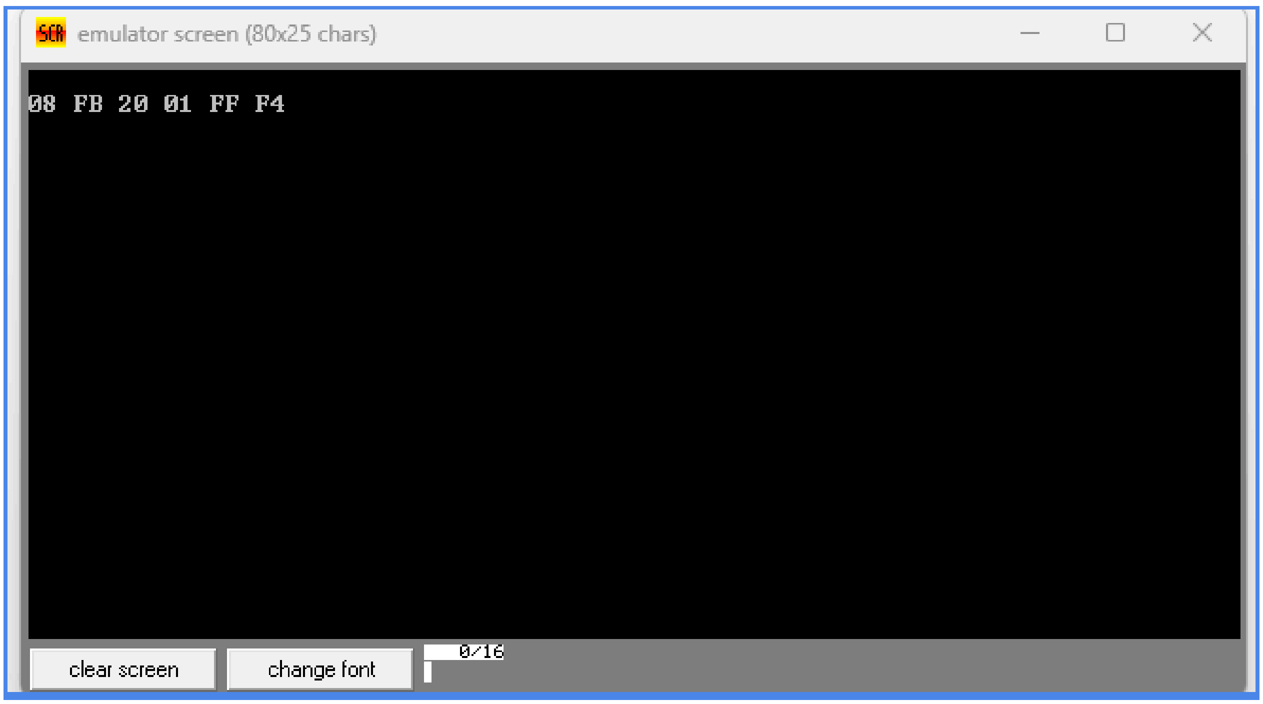
```
digit2:
    add al, 30h
    mov dl, al
    mov ah, 2
    int 21h

    ret

print_hex endp

end main
```

## Output Screenshot:



## Observation Table:

| Num | Instruction | Operation Description | Result(Hex) |
|---|---|---|---|
| 18h | AND 0Fh | Performs bitwise AND between 18h and 0Fh | 08h |
| 2Ah | OR D1h | Performs bitwise **OR** between 2Ah and D1h | FBh |
| C4h | SHL, 3 | Shifts the bits of C4h left by 3 positions | 20h |
| 6Bh | SHR, 6 | Shifts the bits of 6Bh right by 6 positions | 01h |
| FFh | ROL, 2 | Rotates the bits of FFh left by 2 positions | FFh |
| 7Ah | ROR, 7 | Rotates the bits of 7Ah right by 7 positions | F4h |

## I learned:

Through this assignment, I learned how to perform logical, shift, and rotate operations using 8086 Assembly Language.

- Learned how to perform bitwise AND and OR operations on 8-bit data.
- Learned how the SHL and SHR instructions shift binary bits to the left and right.
- Learned how the ROL and ROR instructions rotate bits within an 8-bit value.
- Learned how hexadecimal values represent binary data in Assembly Language.
- Learned how to use registers such as AL and CL for performing different operations.
- Learned how logical and bit manipulation instructions affect the binary representation of data.
- Learned how to observe, calculate, and verify the hexadecimal results of different 8086 instructions.

## Conclusion

In this assignment, I successfully performed logical, shift, and rotate operations using 8086 Assembly Language. The results helped me understand how these instructions manipulate data at the bit level and how hexadecimal values represent the results.